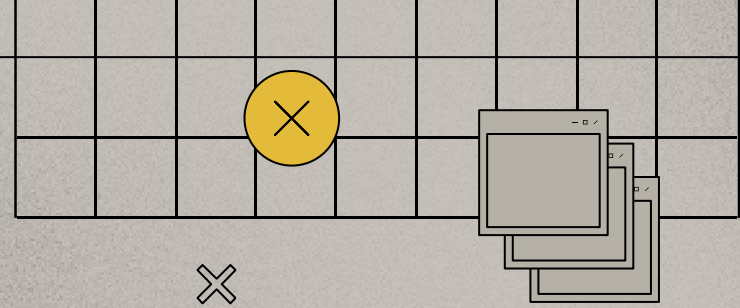
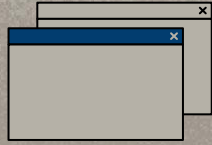
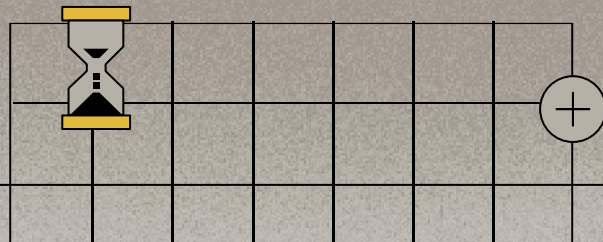




The Device Dilemma: How Tech Shapes Student Success



Team Debsa:
Debanjana HALDAR, Mahsa NIAZI



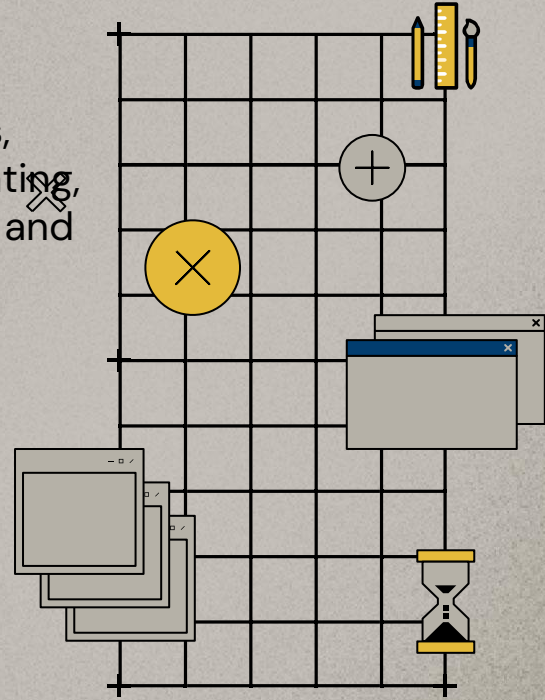
✕ Digital divide (inequality in ability of usage of tech)

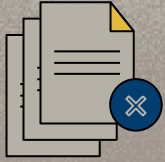
ABOUT

Examining how different devices (paper, tablets, laptops, smartphones) and study methods (note-taking, highlighting, digital tools) affect students' comprehension, retention, and engagement.

WHY

- Educational Equity
- Digital Literacy
- Environmental Impact
- Mental Health and Well-being
- Workplace Preparedness
- Inclusion and Accessibility

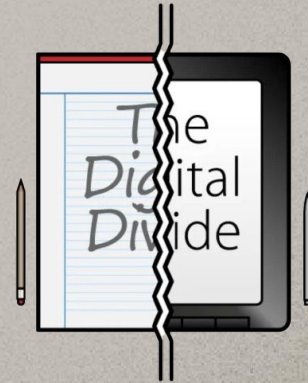


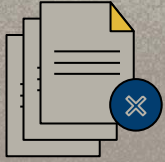


Question 1



How does taking notes with digital devices such as tablets and laptops, as opposed to paper, impact university students' learning and revision processes?



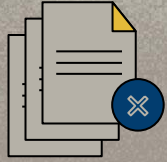


Question 2



How do different combinations of device type, study technique (e.g., handwritten notes, typed notes, highlighting, digital annotation), and learning environment (e.g., classroom, library, home) influence students' comprehension, retention, and learning across different academic subjects?





Question 3



How does the presence and type of distractions (e.g., social media notifications, non-educational websites, multitasking) when using devices for learning purposes impact students' focus, attention span, and overall academic performance?



Team DebSa

Debanjana HALDAR

Mahsa NAIZI

Impact of Note-Taking Methods on Academic Performance

1. Datasets

- Dataset 1: "Note-taking by university students on paper or a computer: Strategies during initial note-taking and revision".

Link to [article's website](#) and its [dataset](#).

This dataset analyzes note-taking strategies used by university students, exploring differences between paper-based and computer-based methods during both initial note-taking and revision phases. It is part of a research study conducted by educational researchers to better understand the impact of technology on learning processes.

Mahsa Niazi found this dataset.

- Dataset 2: "Student Performance Classification Using Artificial Intelligence Techniques"

Link to [article's website](#) and its [dataset](#).

This dataset was found on the UC Irvine Machine Learning Repository, a public collection of datasets used for machine learning research. This dataset was collected via survey from students at the Faculty of Engineering and Faculty of Educational Sciences in Cyprus in 2019. The survey data includes demographics, study habits including note-taking, and various other performance markers.

Debanjana Haldar found this dataset.

2. Selected Dataset

After evaluating the available datasets, we have chosen to utilize the dataset from "Student Performance Classification Using Artificial Intelligence Techniques".

This dataset includes a specific variable on note-taking frequency during classes, which directly aligns with our research focus on the impact of note-taking methods on academic performance. It encompasses a wide range of variables, including student demographics, study habits, and academic performance metrics, allowing for a multifaceted analysis.

Data Source: The dataset is sourced from the UCI Machine Learning Repository, a platform for machine learning datasets.

Data Collection: The data were collected in 2019 from students in the Faculty of Engineering and Faculty of Educational Sciences at Near East University in Cyprus.

Dataset Provider: The dataset is provided by the UC Irvine Machine Learning Repository. The creators of the dataset are Nevriye Yilmaz (nevriye.yilmaz@neu.edu.tr) and Boran Şekeroğlu (boran.sekeroglu@neu.edu.tr) from Near East University in Cyprus.

3. Refinement of Research Questions

We chose to focus on this initial research question:

"How does taking notes with digital devices such as tablets and laptops, as opposed to paper, impact university students' learning and revision processes?"

Based on the selected dataset, we have modified it to:

"How does the frequency of taking notes in classes relate to university students' academic performance, considering factors such as demographics, weekly study hours, and reading frequency over a period of an academic year?"

Justification for Modification:

Data Availability: The chosen dataset includes a variable on the frequency of taking notes in classes but does not specify the medium (paper or digital). Therefore, we adjusted our question to focus on note-taking frequency.

Complementary Variables: The dataset provides additional variables like attendance, weekly study hours, and reading frequency, enabling a comprehensive analysis of factors influencing academic performance.

4. Data Exploration and Cleaning

Upon examining the dataset in OpenRefine, we observed the following:

Data Structure: The dataset comprises 145 instances with 30 features, including categorical variables representing student demographics, study habits, and academic performance indicators.

Missing Values: The dataset documentation indicates no missing values.

Data Consistency: Categorical variables are encoded numerically. We will need to map these numerical codes to their corresponding categorical labels for clarity in analysis.

▼ All	▼ STUDENT ID	▼ 1	▼ 2	▼ 3	▼ 4	▼ 5	▼ 6	▼ 7	▼ 8	▼ 9	▼ 10	▼ 11	▼ 12	▼ 13	▼ 14	▼ 15	▼ 16	▼ 17	▼ 18
★ ↻	1. STUDENT1	2	2	3	3	1	2	2	1	1	1	1	2	3	1	2	5	3	
★ ↻	2. STUDENT2	2	2	3	3	1	2	2	1	1	1	2	3	2	1	2	1	2	
★ ↻	3. STUDENT3	2	2	2	3	2	2	2	2	4	2	2	2	2	1	2	1	2	
★ ↻	4. STUDENT4	1	1	1	3	1	2	1	2	1	2	1	2	5	1	2	1	3	
★ ↻	5. STUDENT5	2	2	1	3	2	2	1	3	1	4	3	3	2	1	2	4	2	
★ ↻	6. STUDENT6	2	2	2	3	2	2	2	2	1	1	3	3	2	1	2	3	1	
★ ↻	7. STUDENT7	1	2	2	4	2	2	2	1	1	3	1	3	1	1	2	4	2	
★ ↻	8. STUDENT8	1	1	2	3	1	1	1	2	2	3	4	3	1	1	4	3	1	
★ ↻	9. STUDENT9	2	1	3	3	2	1	1	1	1	3	2	4	2	1	2	4	1	
★ ↻	10. STUDENT10	2	1	2	3	2	2	1	3	4	2	1	2	3	1	2	3	2	
★ ↻	11. STUDENT11	1	1	1	3	2	2	2	3	2	3	3	4	2	1	3	2	1	
★ ↻	12. STUDENT12	1	1	1	4	1	1	2	4	2	3	5	5	1	1	3	2	3	
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★ ↻	14. STUDENT14	2	1	2	5	2	2	2	1	1	1	2	2	2	1	2	3	1	
★ ↻	15. STUDENT15	3	2	2	4	1	1	2	3	4	2	3	1	2	1	4	1	2	
★ ↻	16. STUDENT16	2	2	2	3	2	2	2	1	1	2	4	4	2	1	3	1	2	
★ ↻	17. STUDENT17	1	1	2	5	2	1	2	1	1	1	2	2	4	1	2	3	2	
★ ↻	18. STUDENT18	2	2	2	2	2	2	2	4	4	4	2	2	2	4	2	4	2	

Data Cleaning Steps:

- Mapping Categorical Variables:** Translate numerical codes to their respective categorical labels as per the dataset documentation.
- Renaming Columns:** Assign meaningful names to columns based on the provided variable descriptions.
- Data Type Conversion:** Ensure all variables are in their appropriate data types
- Handling Inconsistencies:** Verify the consistency of data entries and address any anomalies detected during exploration.

STUDENT ID	Age	Gender	Graduated High-School Type	Scholarship	Additional Work	Artistic/Sports Activity
STUDENT1	22-25	male	Other	50%	Yes	No
STUDENT2	22-25	male	Other	50%	Yes	No
STUDENT3	22-25	male	State	50%	No	No
STUDENT4	18-21	female	Private	50%	Yes	No
STUDENT5	22-25	male	Private	50%	No	No
STUDENT6	22-25	male	State	50%	No	No
STUDENT7	18-21	male	State	75%	No	No
STUDENT8	18-21	female	State	50%	Yes	Yes
STUDENT9	22-25	female	Other	50%	No	Yes
STUDENT10	22-25	female	State	50%	No	No
STUDENT11	18-21	female	Private	50%	No	No
STUDENT12	18-21	female	Private	75%	Yes	Yes
STUDENT13	18-21	female	Private	75%	No	No
STUDENT14	22-25	female	State	100%	No	No

5. Access to Cleaned Dataset

<https://drive.google.com/file/d/1juWiIucfNaVbL8J0ubbf0tuSlZXKmtZn/view?usp=sharing>

6. Additional comments

Given that the Note-taking dataset (the first dataset) primarily contains self-reported data without direct academic performance metrics, we can analyze it to understand students' perceptions and behaviors. While it lacks direct links to academic performance, it provides valuable qualitative insights into students' behaviors, such as their choice of note-taking method (paper, computer, or both), the frequency of distractions during class, and revision strategies like creating review sheets. This qualitative analysis can complement the quantitative findings from the main dataset.

The data was cleaned by rounding numbers where appropriate and ensuring proper data types for analysis. ([Link to cleaned Note-taking dataset](#))